# Design Review

## Synthesis & APR

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# Review Outline

- Synthesis
- APR
  - Clock
  - Reset
  - Reports
- Verification
  - Post-apr simulations (with sdf)
  - Coverage
  - Regression framework (if any)

# Synthesis Overview

- Flow: RTL, constraints, ...
- Design Constraints: clock period, input / output delay, setup / hold skew, ...
- Synthesis Clock & High Fanout Nets
  - *clk_i, rstn_asyn_i, g_clk*
- Reports
  - Clock period: 10 ns (100 MHz)
  - Number of cells:  6896 (comb 6141, seqn 731)
  - Total cell area: 24342 $um^2$ (utilization ratio: 0.6129)
  - Power consumption:  0.87 mW (dynamic: 0.75 mW, leakage: 0.12 mW)
  - Min setup slack: 4.70 ns (REGOUT group, clk_i to sample_clk_o)
    - 7.46 ns (REGIN group, endpoint: pe/south_o_reg[2]/D (DFCNQD1))
  - Min hold slack: 0.01 ns (REGIN group, rstn_async_i to rising-edge removal check against clk)

# APR Overview

- Modes & Scenarios
  - Process Corners (FF 1.1V, SS 0.9V)
  - Timing Derating – Early 0.8, LATE 1.1
  - Floorplan: # of Supertile 12 X12, total area after APR: 172.8 X 172.8 $um^2$
  - Die Utilization : 83.52%
  - APR Clock Period: 10 ns
  - Number of Routing Iterations: 50
  - Minimum Hold Slack : 0.03 (clock path: clk_i -> sample_clk_o)
  - Minimum Setup Slack: 4.73
  - Power Consumption: Static Power: 1.79E-4 W, Dynamic Power 6.15E-4 W

# APR Add On

- Timing and Power Analysis Using Synopsys PrimeTime
- Power analysis: Total Power: 1.029E-3, Dynamic Power 8.606E-4
    - Leakage Power: 1.174E-4

- Timing Analysis: Best case Hold slack : -0.04 (clk_i -> sample_clk_o)
    - Worst case Setup Slack : -0.51

- Block View : Buffer Tree, Clock Tree and Pins

# Post-APR Verification

- External mode ✅
  - Synchronize inputs with sample_clk_o
- BiST mode ✅
  - Synchronize inputs with sample_clk_o
  - Reach stop code in same number of cycles with same signature analyzer output
- Memory mode ✅
  - 1ns delay between memory emulator and matrix multiplier ports
- Output stationary mode ✅
  - 1ns delay between memory emulator and matrix multiplier ports

# Post-APR Verification Status

- What has been verified and what is missing?

| Level | Directed | Edge-case | Random |
|---|---|---|---|
| Matrix Mult | Passed | Zero, Sat | Passed |
| Wrapper (External) | Passed | Zero, Sat | Passed |
| Wrapper (BiST) | Passed | N/A | Passed |
| Wrapper (Memory) | Passed | Zero, Sat | Passed |

- Assertion:
  - Concurrent (check_wb_mem, check_ib_mem, check_ob_mem)
  - immediate (check_done_o)
- Covergroup
  - Memory coverage: input and weight transactions
  - Bins: Zero, Max, Min, Positive, Negative

Note: jump straight into verification testbench, tasks, and results.

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# Backup Slides (Add More if Needed)

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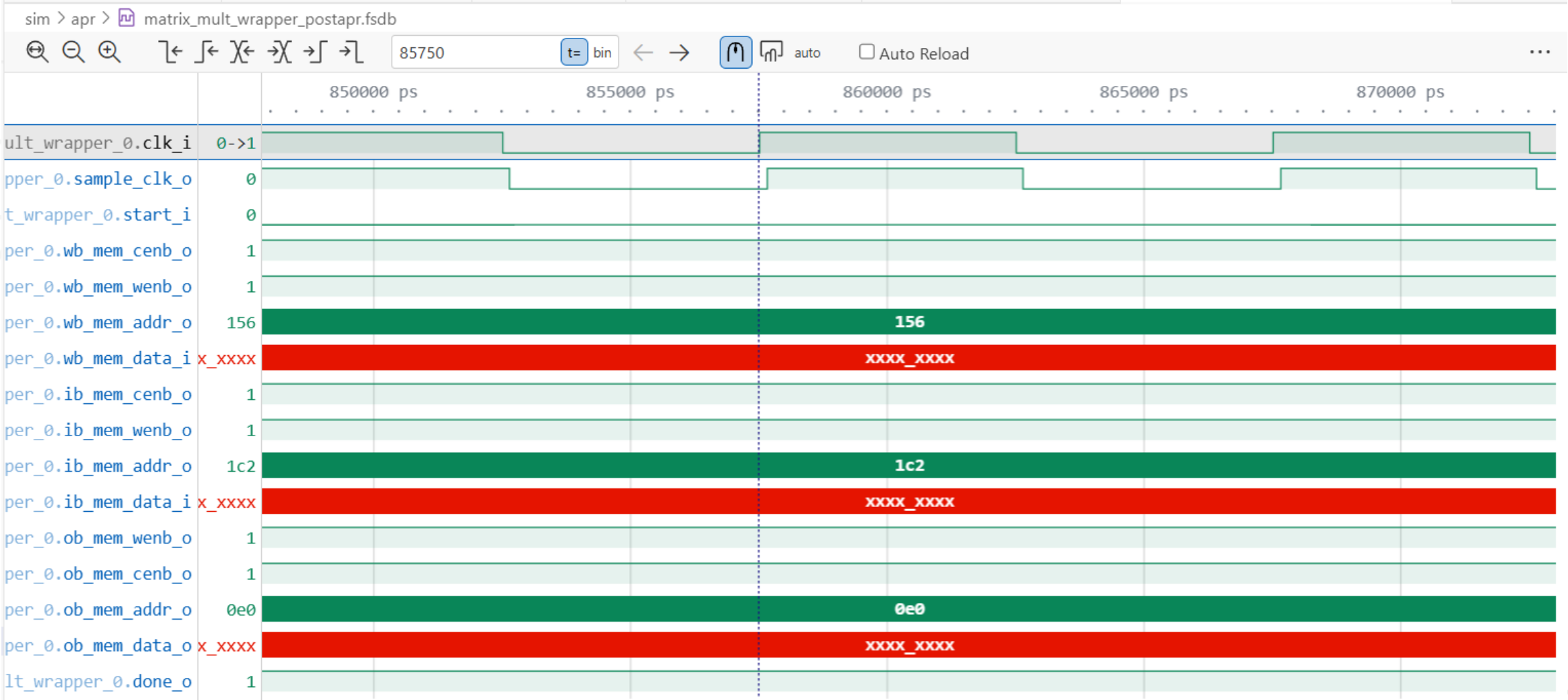
# sample_clk_o vs. clk_i

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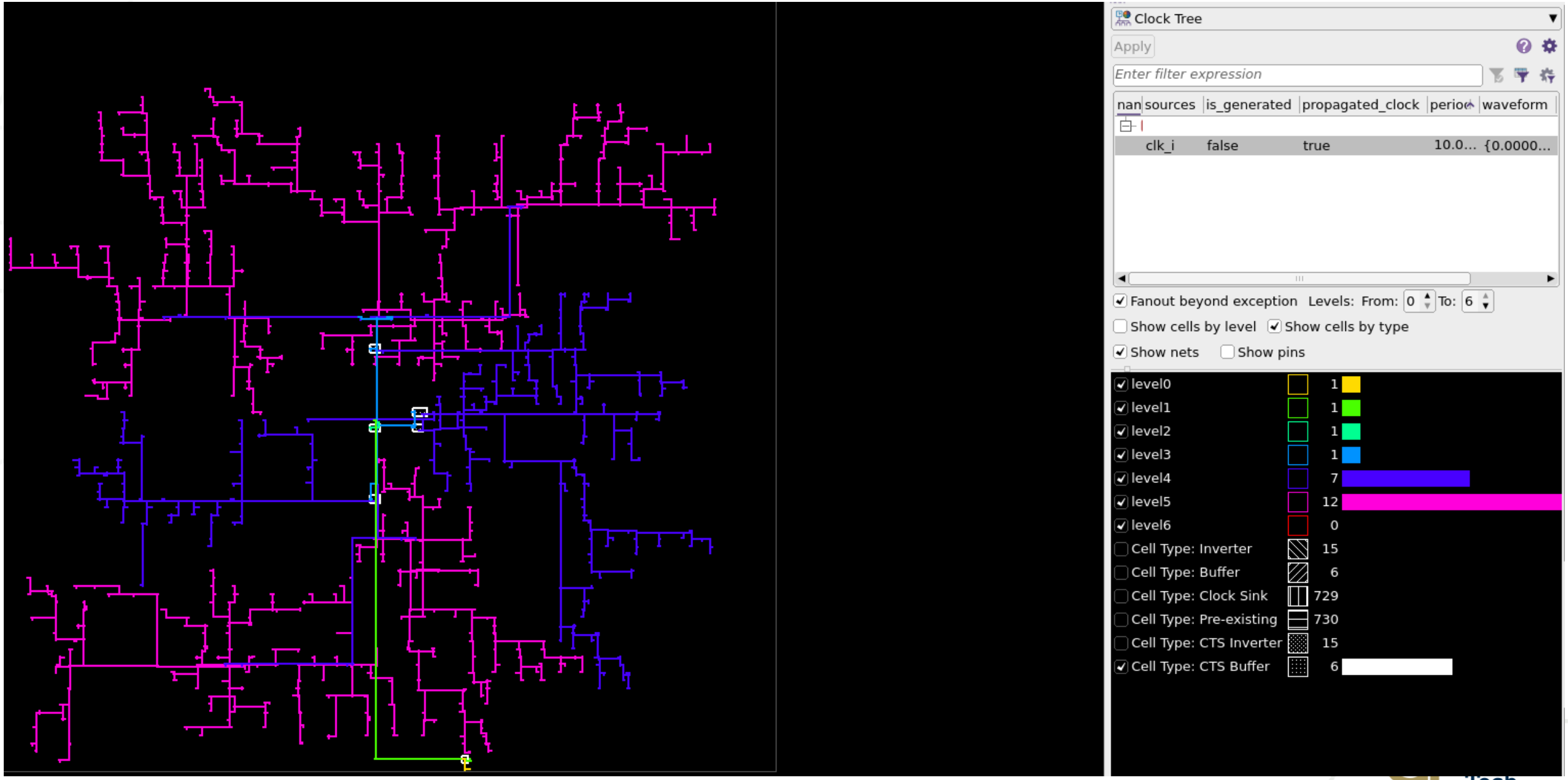
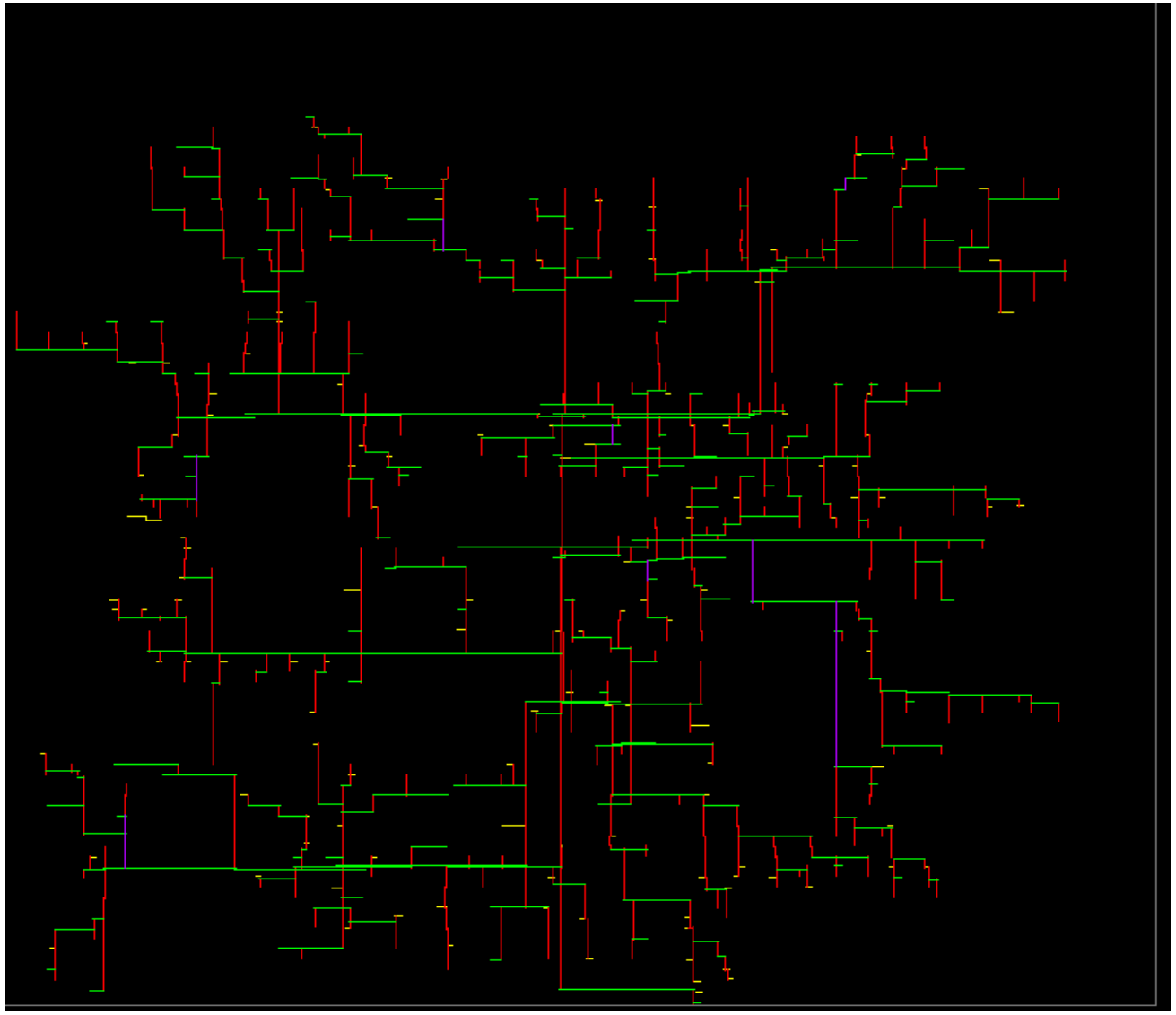
# APR View from ICC2



# Clock Tree View



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# Specifications

| Feature | Description |
|---|---|
| Number of rows | 4 (Programable by Designer) |
| Number of columns | 4 (Programable by Designer) |
| Input data format | Signed Integer (2's Complement) |
| Input data width | 8 bits |
| Output data width | 8 bits |
| Handling data overflow | Saturation |
| Modes supported | Memory, External, & BiST |
| IO ports | See matrix_mult.sv |
| Reset | Active-Low Reset (Reset when 0) |
| Process node | TSMC 65GP |
| Timing model | NLDM |
| Power supply | 0.8 ~ 1 V |
| Target corner | TT |
| Minimum clock frequency | 100 Mhz |
| Highest metal allowed | M7 |
| Target Area | 172.8 um x 172.8 um |

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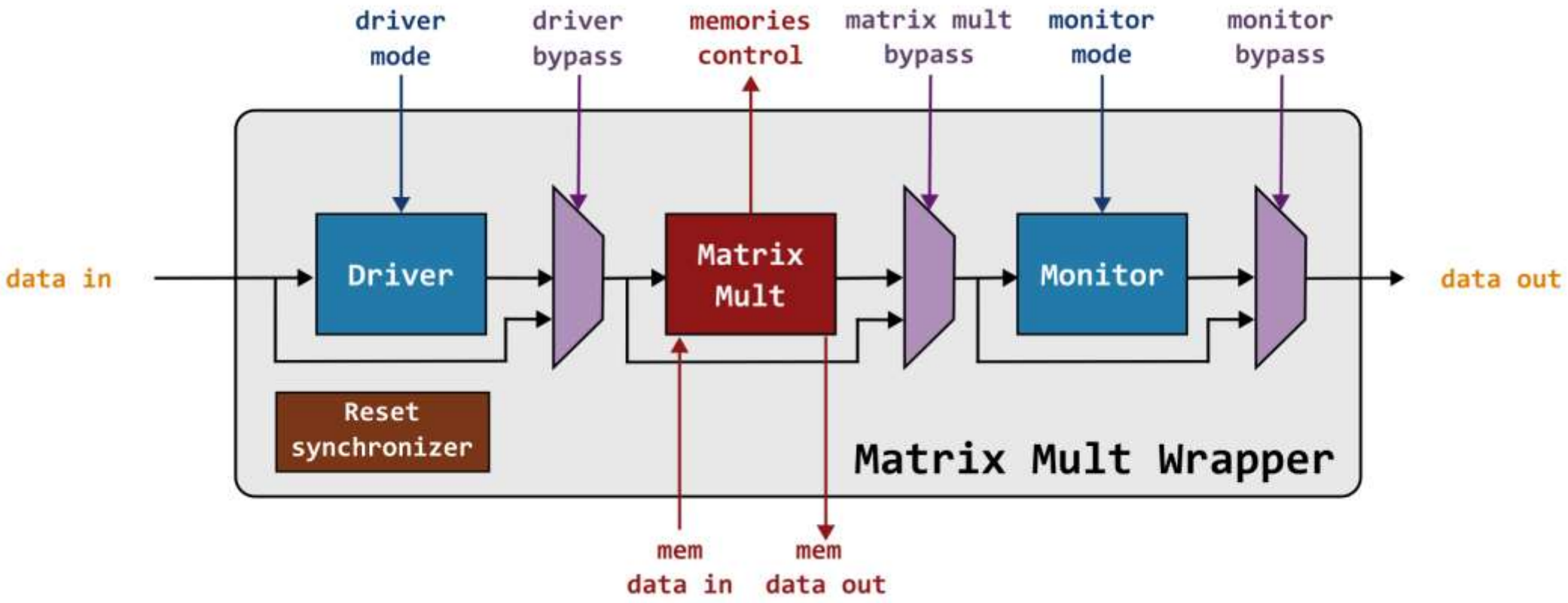
# Matrix Mult Wrapper